

MISUN KIM

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RESEARCH INTERESTS

I study when internal representations in multilingual LLMs support robust behavioural control and when they do not. In my dissertation, transfer success reversed across models and behaviours: each model carried one alignment intervention across languages and failed on the other.

EDUCATION

MSc Speech and Natural Language Processing

Sep 2025 - Sep 2026

University of Sheffield | Sheffield, UK

- Predicted: Distinction
- Dissertation: Cross-Lingual Activation Steering in Large Language Models: Representational Alignment and Behavioural Transfer

BE in Industrial Management Engineering

Mar 2019 - Feb 2024

Hankuk University of Foreign Studies | South Korea

- Double Major: Central Asian Studies (Kazakh language and regional studies) | GPA 3.94/4.5
- First Distinction Scholarship (2023)

RESEARCH EXPERIENCE

MSc Dissertation Research — Reliability of LLM Alignment Interventions

Feb 2026 - Sep 2026

University of Sheffield | Supervisor: Dr Cass Zhao

- Evaluated Contrastive Activation Addition across Qwen3.5-9B, Gemma4-12B-IT, and OLMo3-7B, spanning five language conditions and two behavioural tasks, to test when representation-level interventions transfer reliably across models and settings.
- Identified a representation-behaviour gap: high steering-vector similarity did not reliably predict the magnitude or direction of behavioural transfer across models and tasks, concluding that high representational similarity is not sufficient to guarantee comparable behavioural effects, and leaving the question of necessity unresolved.
- Designed controlled evaluations using bootstrap confidence intervals, Gaussian and permutation random-vector baselines (20 seeds each), matched-layer geometric analysis, and cross-task validation to distinguish robust effects from artefacts.
- Conducted a separate cross-task evaluation using English-derived truthfulness vectors on TruthfulQA, finding that transfer success reversed across models and behaviours: Qwen3.5-9B transferred sycophancy but not truthfulness, OLMo3-7B-Instruct the reverse

Undergraduate Researcher

Mar 2023 - Nov 2023

ICT Mentoring Project, Ministry of Oceans and Fisheries | Supervisor: Gyu-Young Lee (KAIST)

- First-authored LSTM study on container freight forecasting across five universities, reducing RMSE by 71% over the baseline (published ACK 2023)

PUBLICATIONS & MANUSCRIPTS

- Kim, M.*, Kim, Y., Kim, E., Lee, B., Han, Y., & Lee, G. (2023). Implementation of Container Volume Prediction Technology using Deep Learning. ACK Conference Proceedings, 30(2).
- Manuscript in preparation for short paper submission (2026). Cross-lingual and cross-task generalisation of representation-level interventions in multilingual LLMs.

KEY PROJECTS

Post-Training Safety Evaluation Framework

Aug 2026 - Present

- Built a reproducible multilingual LLM safety-evaluation pipeline for SFT, DPO, and RLVR checkpoints, combining typed experiment/result schemas, benchmark-native scoring, deterministic parallel-sample validation, and automated GitHub Actions CI.
- Ran a 50-item parallel EN/KO pilot on OLMo 3 7B using A40 GPUs, finding 2.5–2.9× lower refusal rates for harmful requests in Korean than English across all three post-training stages, motivating larger-scale evaluation of cross-lingual safety consistency.

ESG Compliance Checker

Mar 2026 - Present

- Built an end-to-end compliance verification system over 11 corporate ESG reports, mapping regulatory criteria to retrieved evidence and producing traceable covered/partial/missing judgements using hybrid retrieval and LLM-assisted verification.
- Developed and audited a manually annotated retrieval gold set; hybrid retrieval with reranking improved Hit@5 from 83.3% to 100% and MRR from 0.628 to 0.819 over BM25-only retrieval.
- Used retrieval and verifier error analysis to identify annotation gaps and evidence-selection failures, then modularised the system with FastAPI, automated testing, Docker, and a separate Streamlit interface.

SKILLS

Programming: Python, SQL

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn, clustering (k-means, hierarchical)

LLM Alignment & Interpretability: activation steering (CAA), representation analysis, layer-wise interventions, behavioural evaluation, experimental controls, post-training safety evaluation, cross-lingual evaluation, LLM-as-judge

Research Infrastructure: HPC/SLURM, Weights & Biases, Git, Docker, FastAPI, Flask, ChromaDB

WORK EXPERIENCE

Sustainability Research Analyst

Mar 2025 - May 2025

Eon consulting group | South Korea

- Contributed to 170+ page sustainability reports for major public and financial organisations, conducting TNFD climate scenario analysis and developing a CSDDD-aligned supply-chain risk assessment framework.

Field Service Manager (Data & Operations)

Mar 2024 - Aug 2024

Payhere | South Korea

- Built workflow automation using the Slack API, Google Sheets, and Looker Studio to analyse 10,000+ service records, reducing unnecessary field visits by 17%.

TEACHING EXPERIENCE

Language Tutor (Private)

Mar 2021 - Nov 2025

- Provided individualized Korean/English instruction to 5+ learners aged 6–18, from early literacy to CSAT-level advanced reading

LANGUAGES

Korean (Native), English (Fluent), Spanish (Intermediate), Kazakh (Intermediate)